

RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES, BASAR

(AY 2026-27)

Computer Science & Engineering

Subject: Computer Networks

Subject Code: CS3701

Date: 29/09/2026

Time: 3 Hrs

Exam: Lab Internals

Max Marks: 30

1Q. Construct a Tree Topology in Cisco Packet Tracer and incorporate a Hybrid Network Topology within it by connecting a Ring Topology and a Star Topology through a router to form an interconnected network. Use a total of 8 PCs and connect a server to the network. Configure all networking devices with appropriate IP addresses, subnet masks, and default gateways. Perform the following tasks:

- a)** Configure the server as an **Email Server** and demonstrate email communication between two PCs.
- b)** Perform a **ping test between two PCs within the same network** and verify successful communication.
- c)** Use **Simulation Mode** to simulate communication between two PCs belonging to **different networks** through the router. Observe the movement of the packets and record the results.
- d)** Perform **ARP (Address Resolution Protocol)** on a PC and display the ARP table. Identify the corresponding IP and MAC addresses and explain the ARP process.

E)Network Diagram

2Q. Construct a hybrid network topology in Cisco Packet Tracer by connecting a ring topology with four star topologies. Connect a total of 12 PCs to the four star networks and use suitable switches to connect the PCs. Configure the switches and assign appropriate IP addresses to all PCs. Represent the different network sections using different colors. Perform the following tasks:

- a) Perform the necessary **switch configuration**, including basic switch settings, device names, and appropriate interface configuration.
- b) Assign suitable IP addresses and subnet masks to all PCs. Perform a **ping test between two PCs connected to the same star topology** and verify successful communication.
- c) Use **Simulation Mode** in Cisco Packet Tracer to simulate a message/packet from one PC to another PC within the same star topology. Observe the packet flow through the switch and record the results.
- d) Perform an **ARP request** from a PC to another PC in the same star topology. Display the ARP table and identify the corresponding IP and MAC addresses. Explain how ARP resolves the destination IP address into a MAC address.

E) Network Diagram

3Q. Construct and configure a WAN in Cisco Packet Tracer consisting of three branches connected through suitable routers. Design the complete network in a tree topology and use different IP address classes for each branch: Class A for Branch 1, Class B for Branch 2, and Class C for Branch 3. Represent the three branches using different colors and connect a server to the network. Configure the network and perform the following tasks:

- a) Configure the server as an **Email Server** and demonstrate email communication between users of the network.
- b) Configure the required IP addresses, subnet masks, default gateways, and router interfaces, and perform a **ping test from a PC in Branch 1 to a PC in Branch 3**. Verify successful communication across the WAN.
- c) Use **Simulation Mode** in Cisco Packet Tracer to simulate a message/packet travelling from a device in **Branch 1 to a device in another branch**. Observe the path followed by the packet through the routers and record the results.
- d) Perform **ARP** on a PC in the network. Display the ARP table and identify the corresponding IP addresses and MAC addresses. Explain the role of ARP during communication.
- e) Draw the complete WAN tree topology, configure all routers and end devices, use different colors to identify the three branches, and record the IP addressing scheme, configurations, observations, and results.

